EEE-101 - Control Systems Engineering

Tutorial - 3 - Answers

1. $\dfrac{X_1(s)}{F(s)} = \dfrac{s+1}{s^4+5s^3+8s^2+5s+1}$.

2. $\dfrac{\theta_2(s)}{T(s)} = \dfrac{s^2+9s+8}{s^4+11s^3+33s^2+83s+24}$.

3. Approach-1)

Given: Output $y(t) = v_o(t)$.

Let state variables:

$$\begin{aligned} x_1(t) &= v_{c1}(t), \\ x_2(t) &= i_L(t), \\ x_3(t) &= v_o(t) = v_{c2}(t). \end{aligned}$$

State-space representation:

$$\begin{aligned} \dot{x} &= \begin{bmatrix} -\frac{1}{RC_1} & -\frac{1}{C_1} & -\frac{1}{RC_1} \\ -\frac{1}{L} & 0 & 0 \\ -\frac{1}{RC_2} & 0 & -\frac{1}{RC_2} \end{bmatrix} x + \begin{bmatrix} \frac{1}{RC_1} \\ \frac{1}{L} \\ \frac{1}{RC_2} \end{bmatrix} v_i(t) \\ y &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x\cdot \end{aligned}$$

Approach-2)

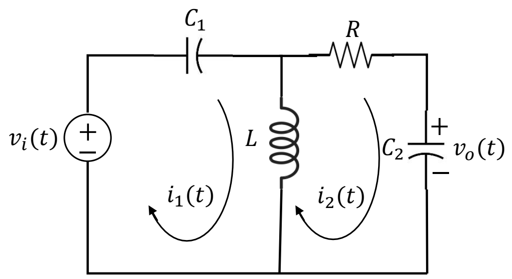

Figure 1

Given: Output $y(t) = v_o(t) = \dfrac{1}{C_2}\int i_2(t)\,\mathrm{d}t$.

Let state variables:

$$\begin{aligned} x_1(t) &= \int \left(\int \left(\int i_2(t)\,\mathrm{d}t \right)\,\mathrm{d}t \right)\,\mathrm{d}t, \\ x_2(t) &= \int \left(\int i_2(t)\,\mathrm{d}t \right)\,\mathrm{d}t, \\ x_3(t) &= \int i_2(t)\,\mathrm{d}t. \end{aligned}$$

State-space representation:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\dfrac{1}{LC_1C_2R} & -\dfrac{1}{LC_1} & -\dfrac{1}{C_2R} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{R} \end{bmatrix} v_i(t)$$
$$y = \begin{bmatrix} 0 & 0 & \dfrac{1}{C_2} \end{bmatrix} x.$$

4. Let state variables:

$$x_1(t) = y(t),$$
$$x_2(t) = \dot{y}(t),$$
$$x_3(t) = \ddot{y}(t).$$

State-space representation:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -6 & -4 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u(t),$$
$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}.$$

5.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u,$$
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Alternative solution:

$$\dot{x}_1 = -2x_1 + u,$$
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x_1.$$

However, reducing the second state may lead to a loss of generality. Therefore, we will consider the initial representation as the preferred solution for this problem.

6.

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} \dfrac{s-1}{s^2+s+6.5} & \dfrac{s}{s^2+s+6.5} \\ \dfrac{s+7.5}{s^2+s+6.5} & \dfrac{6.5}{s^2+s+6.5} \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}$$

7. $y(t) = t + 5e^{-3t}, \quad t \geq 0.$

9.

$$x(t) = \begin{bmatrix} -2.5e^{-t} + 5e^{-2t} - 1.5e^{-3t} \\ 2.5e^{-t} - 10e^{-2t} + 4.5e^{-3t} \\ -2.5e^{-t} + 20e^{-2t} - 13.5e^{-3t} \end{bmatrix}$$